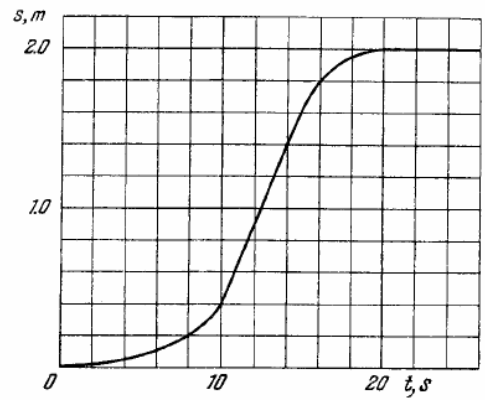
PHY101: Introduction to Physics-1

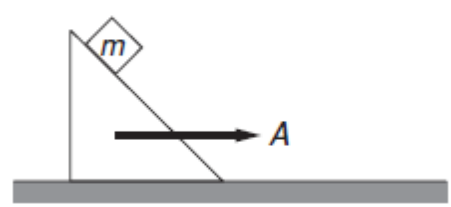
Monsoon, 2017

Tutorial-4

1. A point moves rectilinearly in one direction. In the graph, the distance s traversed by the point as a function of time t is shown. Using the plot find the average velocity of the point during the time of motion. Also, indicate the region of maximum velocity in the graph and calculate the value.



2. A particle moves in a plane with constant radial velocity $\dot{r} = 4\ m/s$, starting from the origin. The angular velocity is constant and has magnitude $\dot{\theta} = 2\ rad/s$. When the particle is 3 m from the origin, find the magnitude of (a) the velocity and (b) the acceleration.

3. A 5kg mass moves under the influence of a force $\vec{F} = (4t^2\hat{\imath} - 3t\hat{\jmath})N$, where t is the time in seconds. It starts at rest from the origin at $t = 0$. Find: (a) its velocity ($\vec{v}$); (b) its position ($\vec{r}$); and (c) $\vec{r}$ x $\vec{v}$, for any later time.

4. A 45° wedge is pushed along a table with constant acceleration a. A block of mass m slides without friction on the wedge. Find the block's acceleration. Gravity is directed down.